Robotics and Its Application AI352

Unit 7: AI in Robotics

Topics Cover: Introduction to machine learning in Robotics, Reinforcement learning for robot control and navigation, Natural language processing for human robot interaction, Ethical and Societal implications of AI in robotics

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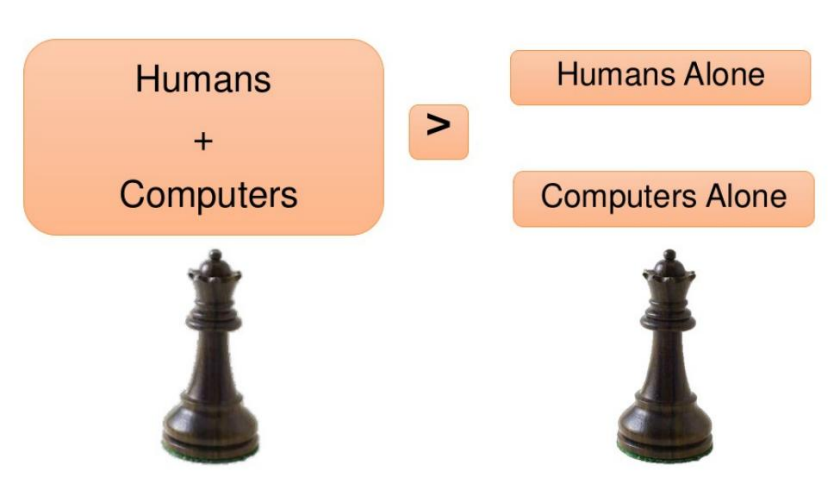
Artificial Intelligence

- Artificial intelligence (AI) refers to the simulation of human intelligence in machines that are programmed to think like humans and mimic their actions.
- The term may also be applied to any machine that exhibits traits associated with a human mind such as learning and problem-solving.
- Artificial intelligence is based on the principle that human intelligence can be defined in a way that a machine can easily mimic it and execute tasks, from the most simple to those that are even more complex.
- The goals of artificial intelligence include mimicking human cognitive activity.



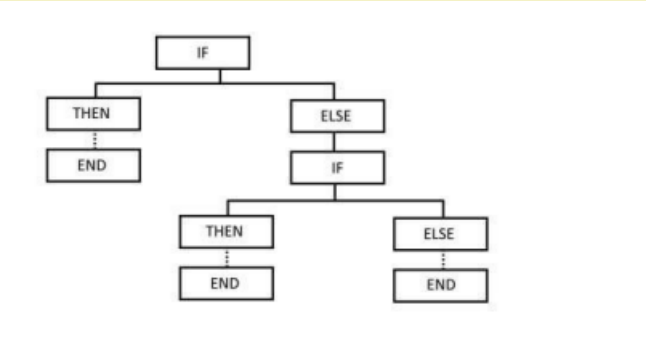
Types

- “Strong” Artificial Intelligence
  - Computers thinking at a level that meets or surpasses people
  - Computers engaging in abstract reasoning & thinking
  - This is not what we have today
  - There is no evidence that we are close to Strong AI
- “Weak” Pattern-Based Artificial Intelligence
  - Computers solve problems by detecting useful patterns
  - Pattern-based AI is an Extremely powerful tool
  - Has been used to automate many processes today
  - Driving, language translation
  - This is the dominant mode of AI today

Major AI Approaches

- Two Major AI Techniques

– Logic and Rules-Based Approach



– Machine Learning (Pattern-Based Approach)

Rules based approach

- Logic and Rules-Based Approach

– Representing processes or systems using logical rules

– Top-down rules are created for computer

– Computers reason about those rules

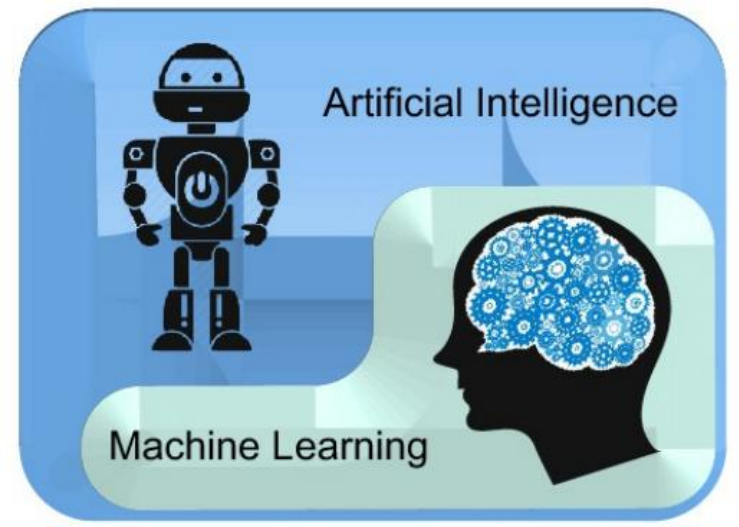
– Can be used to automate processes

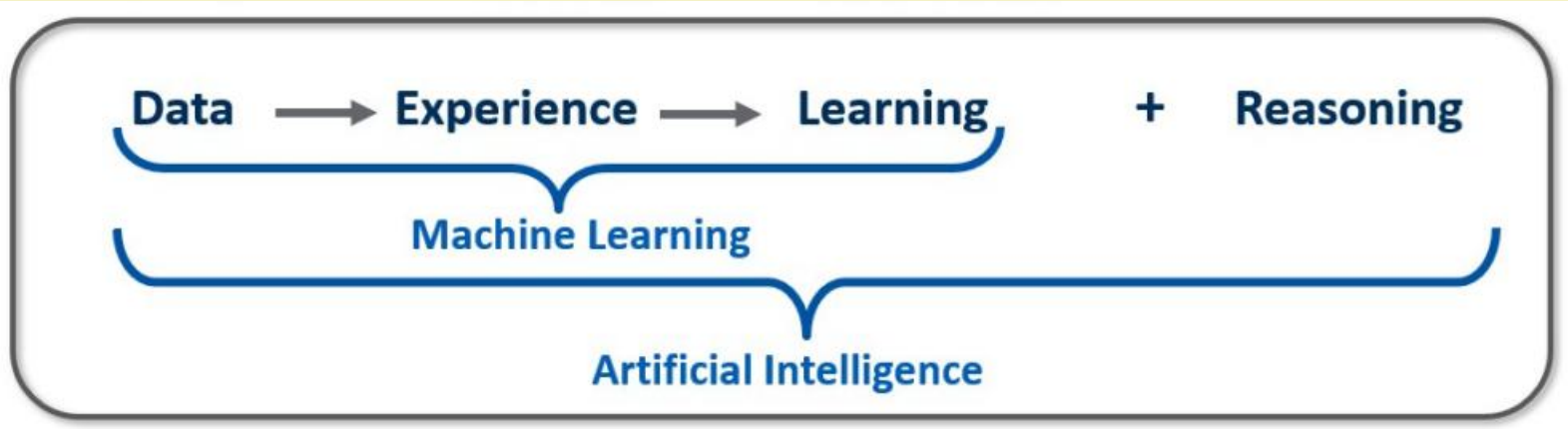
– Example within law – Expert Systems

- Turbotax
- Personal income tax laws
- Represented as logical computer rules
- Software computes tax liability

Learning vs. Designing

- AI is a bigger concept to design intelligent machines that can simulate human thinking capability and behavior, whereas, machine learning is an application or subset of AI that allows machines to learn from data without being programmed explicitly.



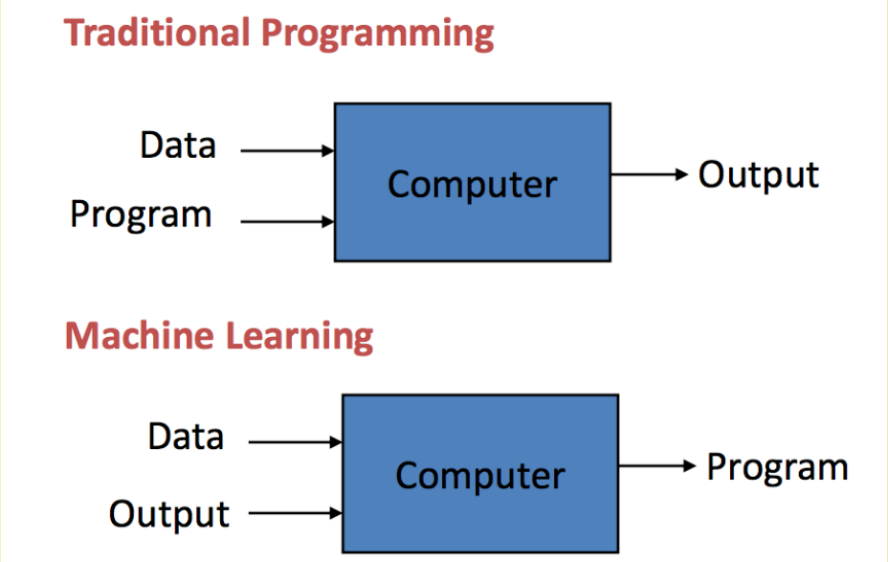


Machine Learning

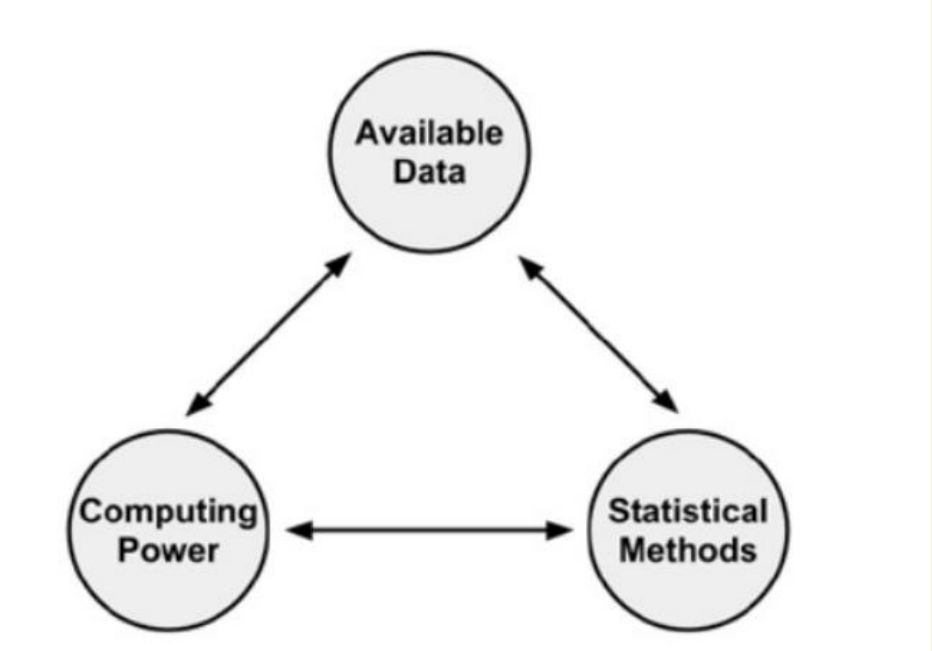
- Machine learning is an application of artificial intelligence (AI) that provides systems the ability to automatically learn and improve from experience without being explicitly programmed. Machine learning focuses on the development of computer programs that can access data and use it learn for themselves.
- The process of learning begins with observations or data, such as examples, direct experience, or instruction, in order to look for patterns in data and make better decisions in the future based on the examples that is provided.
- The primary aim is to allow the computers learn automatically without human intervention or assistance and adjust actions accordingly.

Origins of Machine Learning

- The earliest databases recorded information from the observable environment.
- Astronomers recorded patterns of planets and stars; biologists noted results from experiments crossbreeding plants and animals; and cities recorded tax payments, disease outbreaks, and populations. Each of these required a human being to first observe and second, record the observation.
- Today, such observations are increasingly automated and recorded systematically in evergrowing computerized databases



- The field of study interested in the development of computer algorithms for transforming data into intelligent action is known as machine learning.



Uses and Abuses

- Predict the outcomes of elections
- Identify and filter spam messages from e-mail
- Foresee criminal activity
- Automate traffic signals according to road conditions
- Produce financial estimates of storms and natural disasters
- Examine customer churn
- Create auto-piloting planes and auto-driving cars
- Stock market predition
- Target advertising to specific types of consumers

Recognizing patterns

- Pattern recognition is the automated recognition of patterns and regularities in data. It has applications in

– statistical data analysis,

– signal processing,

– image analysis,

– information retrieval,

– bioinformatics,

– data compression,

– computer graphics and

– machine learning.

How do machine learn ?

- A commonly cited formal definition of machine learning, proposed by computer scientist Tom M. Mitchell, says that a machine is said to learn if it is able to take experience and utilize it such that its performance improves up on similar experiences in the future.
- This definition is fairly exact, yet says little about how machine learning techniques actually learn to transform data into actionable knowledge.

Training a dataset

- The process of fitting a particular model to a dataset is known as training.
- Why is this not called learning? First, note that the learning process does not end with the step of data abstraction.
- Learning requires an additional step to generalize the knowledge to future data.
- Second, the term training more accurately describes the actual process undertaken when the model is fitted to the data.

Process of ML

Practical Machine Learning

Inputs

Output

X	Y	Z
5	2	14
8	5	22
4	8	14
9	2	20
7	1	15
7	8	23

Z = ? ---> ML Model

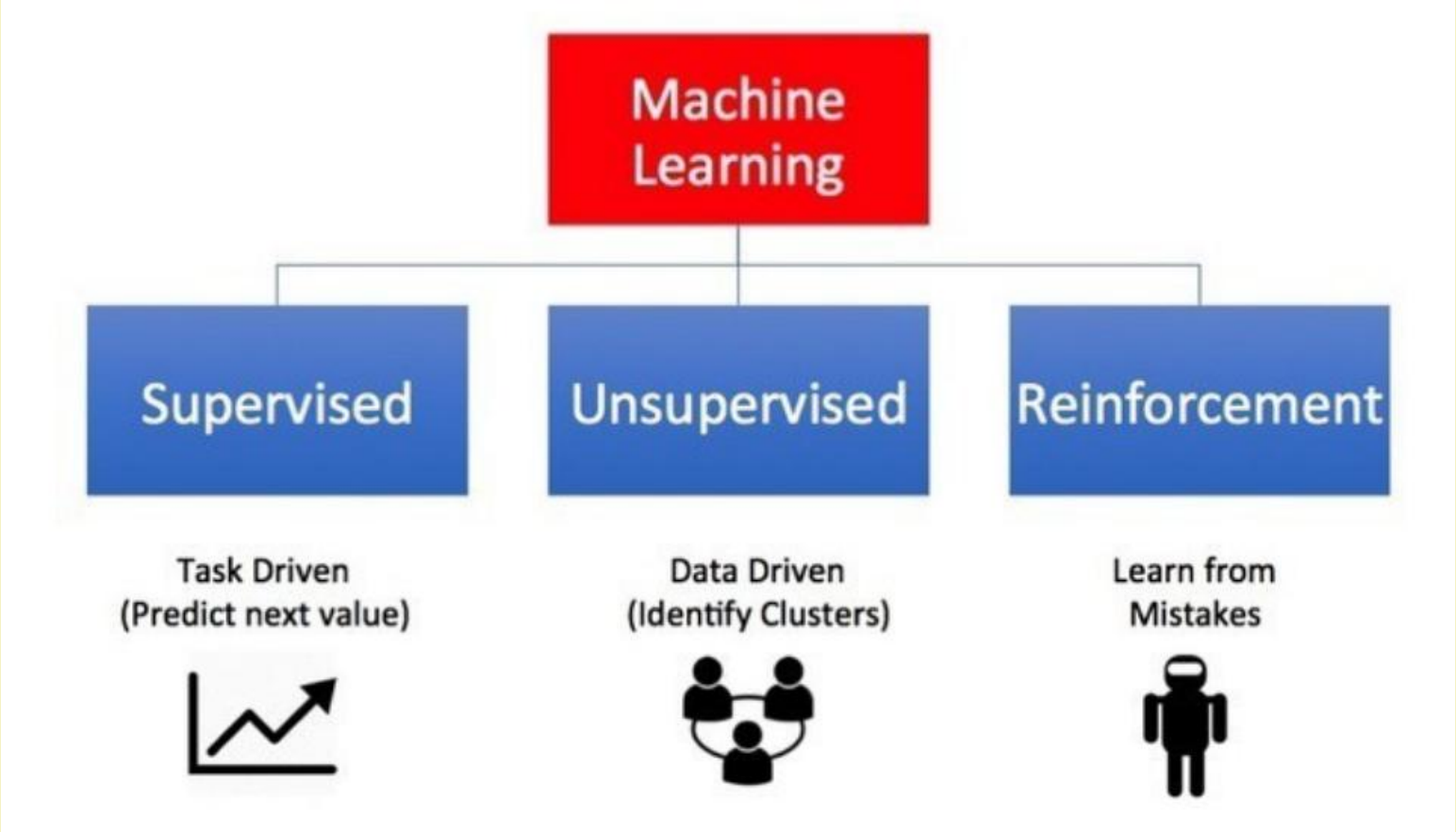
X	Y	Z	Pre	Error
5	2	14	12	-2
8	5	22	21	-1
4	8	14	16	+2
9	2	20	20	0
7	1	15	15	0
7	8	23	22	-1

Z = 2X + Y ---> ML Model

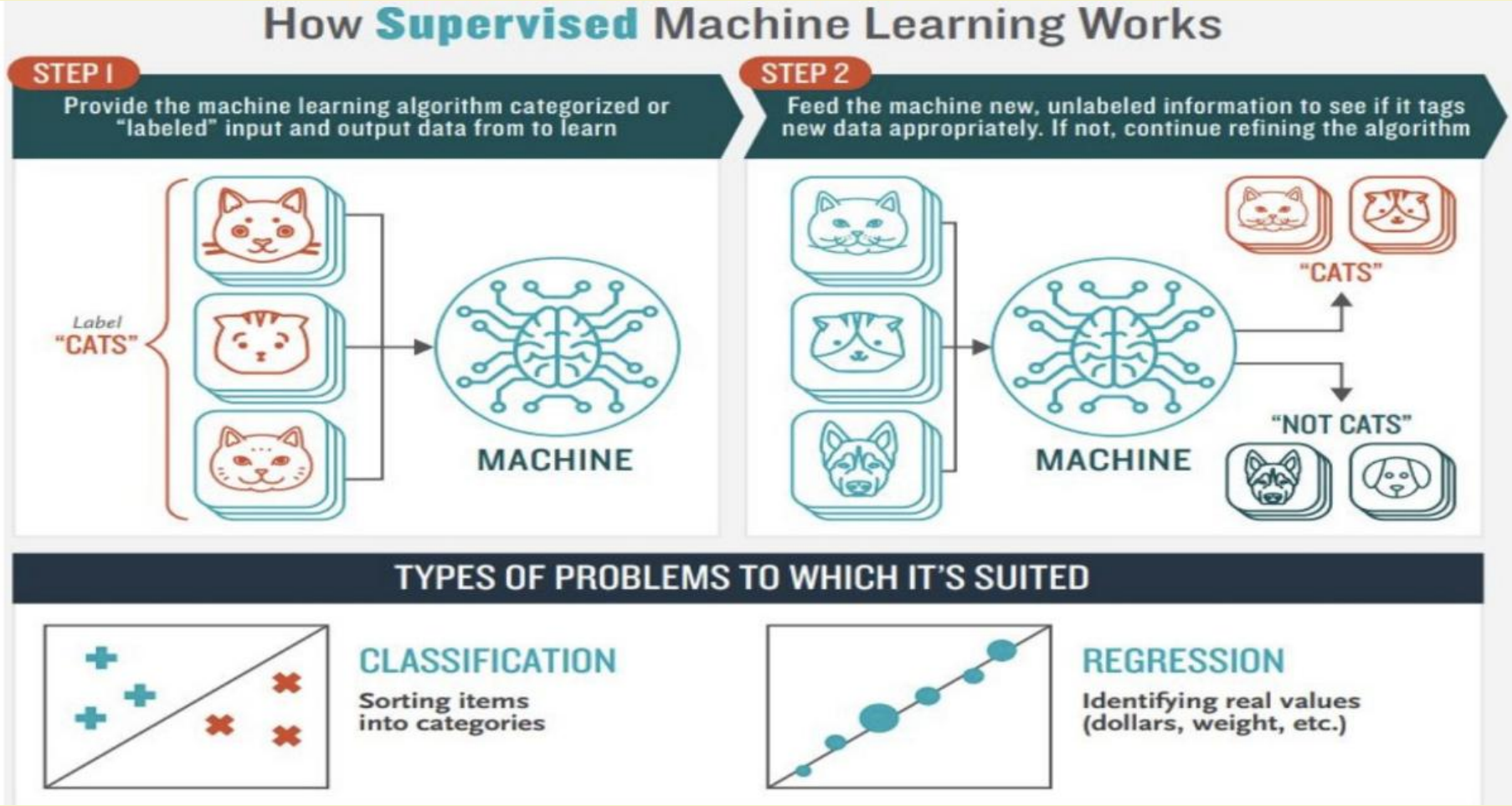
Prediction ---> X = 6 Y = 8 Z = ?

if 20 == 19: 95%

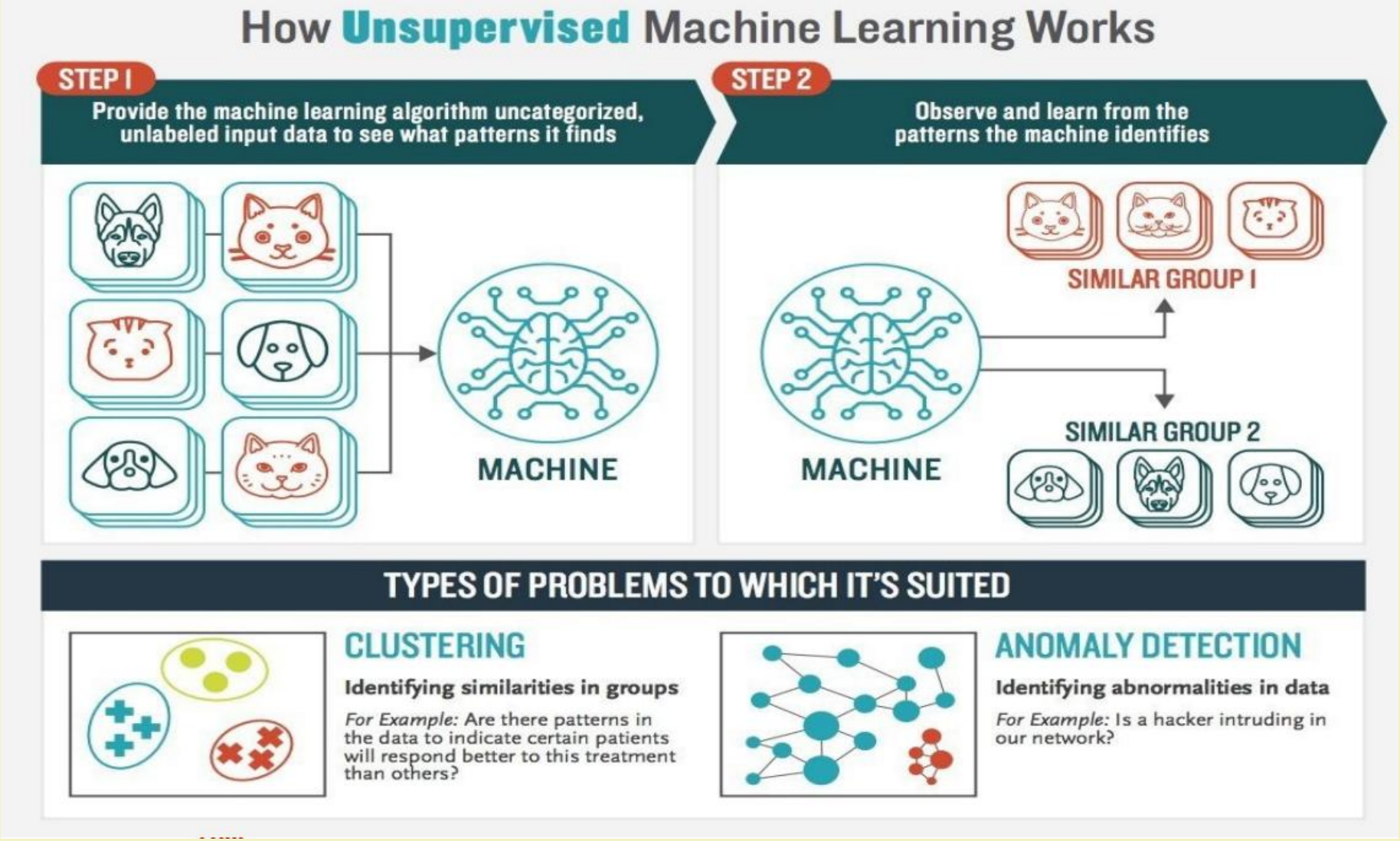
Types of Machine Learning



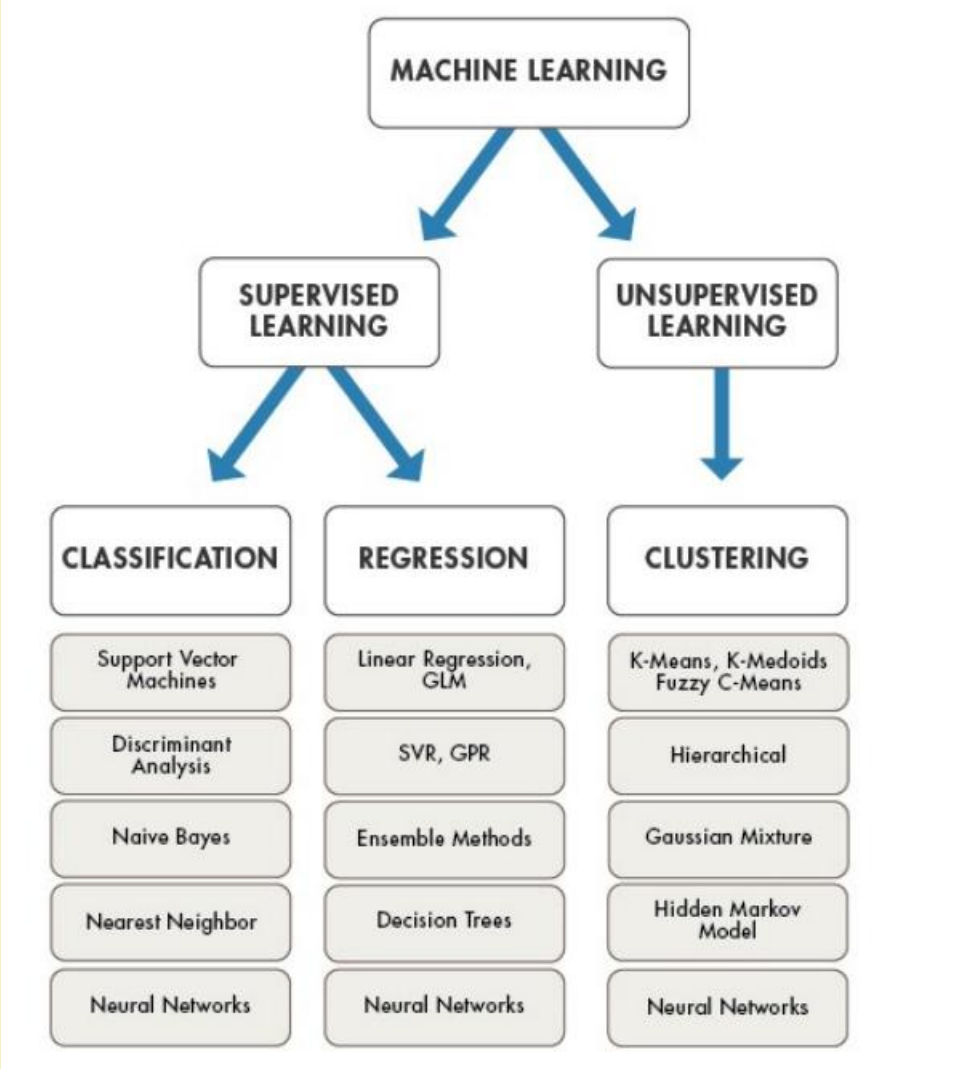
Supervised Machine Learning



Unsupervised Machine Learning



Typical Algorithms in Machine Learning



Intelligent Robots

- You can give it different names, an intelligent software module, an algorithm, a machine learning model, or a computer program that makes the intelligent decision for the integrated hardware interface function, without being explicitly programmed."
- Example:

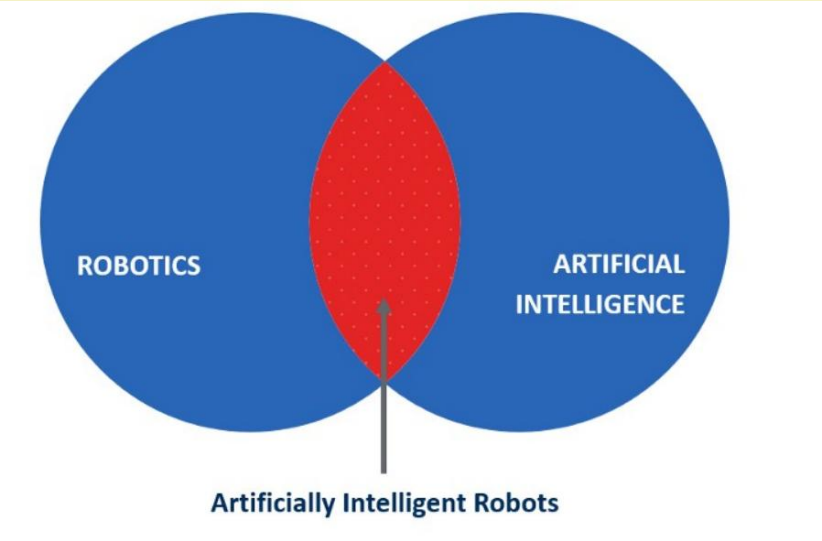
1- Tesla — is an excellent example of an intelligent robot that makes dynamic decisions based on cognitive power and leverages the learning and experience infused by a software program.

2- Drones — are autonomous and AI-based robots. They can mimic human intelligence in navigating their way to the target. They can also do all kinds of things without ever being programmed to do specific static actions.

3- Alexa — is another robot that performs specific actions based on human voice commands. Alexa provides an API (Application Programming Interface) based platform to integrate with any other pluggable interface.

Scope of ML in Robotics

- Vision – AI is helping robots detect items they've never seen before and recognize objects with far greater detail.
- Grasping – robots are also grasping items they've never seen before with AI and machine learning helping them determine the best position and orientation to grasp an object.
- Motion Control – machine learning helps robots with dynamic interaction and obstacle avoidance to maintain productivity.
- Data – AI and machine learning both help robots understand physical and logistical data patterns to be proactive and act accordingly.



ML & Robotics: Computer Vision

- Though related, some would argue the correct term is machine vision or robot vision rather than computer vision, because “robots seeing” involves more than just computer algorithms; engineers and roboticists also have to account for camera hardware that allow robots to process physical data.
- Robot vision is very closely linked to machine vision, which can be given credit for the emergence of robot guidance and automatic inspection systems.
- The slight difference between the two may be in kinematics as applied to robot vision, which encompasses reference frame calibration and a robot’s ability to physically affect its environment.

ML & Robotics: Imitation Learning

- Imitation learning is closely related to observational learning, a behavior exhibited by infants and toddlers.
- Imitation learning is also an umbrella category for reinforcement learning, or the challenge of getting an agent to act in the world so as to maximize its rewards.
- Bayesian or probabilistic models are a common feature of this machine learning approach.
- The question of whether imitation learning could be used for humanoid-like robots was postulated as far back as 1999.

ML & Robotics: Self Supervised Learning

- Self-supervised learning approaches enable robots to generate their own training examples in order to improve performance; this includes using a priori training and data captured close range to interpret “long-range ambiguous sensor data.”
- It’s been incorporated into robots and optical devices that can detect and reject objects (dust and snow, for example); identify vegetables and obstacles in rough terrain; and in 3D-scene analysis and modeling vehicle dynamics

ML & Robotics: Self Supervised Learning

- Watch-Bot is a concrete example, created by researchers from Cornell and Stanford, that uses a 3D sensor (a Kinect), a camera, laptop and laser pointer to detect 'normal human activity', which are patterns that it learns through probabilistic methods.
- Watch-Bot uses a laser pointer to target the object as a reminder (for example, the milk that was left out of the fridge).
- In initial tests, the bot was able to successfully remind humans 60 percent of time (it has no conception of what it's doing or why), and the researchers expanded trials by allowing its robot to learn from online videos (called project RoboWatch).
- Other examples of self-supervised learning methods applied in robotics include a road detection algorithm in a front-view monocular camera with a road probabilistic distribution model (RPDM) and fuzzy support vector machines (FSVMs), designed at MIT for autonomous vehicles and other mobile on-road robots.
- Autonomous learning, which is a variant of selfsupervised learning involving deep learning and unsupervised methods, has also been applied to robot and control tasks.

ML & Robotics: Medical Technologies

- An assistive robot (according to Stanford's David L. Jaffe) is a device that can sense, process sensory information, and perform actions that benefit people with disabilities and seniors (though smart assistive technologies also exist for the general population, such as driver assistance tools). Movement therapy robots provide a diagnostic or therapeutic benefit.
- Both of these are technologies that are largely (and unfortunately) still confined to the lab, as they're still cost-prohibitive for most hospitals in the U.S. and abroad.
- Early examples of assistive technologies included the DeVAR, or desktop vocational assistant robot, developed in the early 1990s by Stanford and the Palo Alto Veterans Affairs Rehabilitation Research and Development.
- More recent examples of machine learning-based robotic assistive technologies are being developed that include combining assistive machines with more autonomy, such as the MICO robotic arm (developed at Northwester University) that observes the world through a Kinect Sensor.

ML & Robotics: Multi Agent Learning

- Coordination and negotiation are key components of multi-agent learning, which involves machine learning-based robots (or agents – this technique has been widely applied to games) that are able to adapt to a shifting landscape of other robots/agents and find "equilibrium strategies."
- Examples of multi-agent learning approaches include no-regret learning tools, which involve weighted algorithms that "boost" learning outcomes in multi-agent planning, and learning in market-based, distributed control systems.
- A more concrete example is an algorithm for distributed agents or robots created by researchers from MIT's Lab for Information and Decision Systems in late 2014.
- Robots collaborated to build a better and more inclusive learning model than could be done with one robot (smaller chunks of information processed and then combined), based on the concept of exploring a building and its room layouts and autonomously building a knowledge base.

ML & Robotics: Bayesian Models

- Bayesian models – Method of statistical inference that casts statistical problems in the framework of decision making.
- It entails formulating subjective prior probabilities to express pre-existing information, careful modeling of the data structure, checking and allowing for uncertainty in model assumptions, formulating a set of possible decisions and a utility function to express how the value of each alternative decision is affected by the unknown model parameters.

ML & Robotics: Inverse Optimal Control

Inverse optimal control – Also known as inverse reinforcement learning, it's the problem of recovering an unknown reward function in a Markov decision process from expert demonstrations of the optimal policy.

ML & Robotics: SVM

Support vector machines – Also called support vector networks, SVMs are supervised learning models with associated learning algorithms that analyze data used for classification and regression analysis.

Reinforcement Learning for Robot Control and Navigation

- In robotics, Reinforcement Learning (RL) is a "learn by doing" approach where a robot (the agent) interacts with its environment to discover the best actions to achieve a goal. Unlike traditional programming, which requires rigid rules, RL allows robots to handle unpredictable, real-world situations like moving through crowds or over rough terrain.
- How RL Works in Robotics
- The robot follows a loop of Action → Observation → Reward:
- The Agent: The robot's "brain" making decisions.
- The State: What the robot "sees" (e.g., sensor data from LiDAR, cameras, or GPS).
- The Action: What the robot does (e.g., "turn left," "increase wheel speed").
- The Reward: A numerical score. Reaching the goal gives a positive reward; hitting an obstacle gives a negative penalty.

RL for Robot Navigation

- Navigation is about moving from Point A to Point B safely. RL is particularly effective for mapless navigation, where the robot doesn't have a pre-loaded floor plan.
- Obstacle Avoidance: Robots learn to interpret "raw" sensor data (like dots on a laser scan) and map them directly to movements that avoid collisions.
- Dynamic Environments: RL helps robots adjust to moving people or changing lighting, which often confuses traditional pathfinding algorithms like A* or Dijkstra.
- Complex Terrains: Four-legged (quadruped) robots use RL to learn energy-efficient gaits for climbing stairs or jumping over gaps.

RL for Robot Control

- Control refers to the low-level management of motors and joints.
- End-to-End Control: Deep RL can map raw camera images directly to joint torques (motor power), skipping the need for manual physics calculations.
- High Precision: Used for tasks like peg-in-hole insertion or object flipping where tiny adjustments are needed.
- Manipulation: Robots learn the "feel" of objects to apply the right amount of pressure when grasping fragile items.

Popular RL Algorithms in Robotics

Algorithm	Best Used For	Why?
PPO (Proximal Policy Optimization)	General Navigation	Stable and easy to tune for smooth movements.
SAC (Soft Actor-Critic) Complex Control	Complex Control	Great for multi-jointed arms as it encourages exploration.
TD3 / DDPG	Precise Motor Tasks	Handles "continuous" actions (like exact speed) very well.
Q-Learning	Simple Navigation	Good for basic "grid" movements and discrete tasks.

Natural language processing

- NLP enables computers and digital devices to recognize, understand and generate text and speech by combining computational linguistics—the rule-based modeling of human language—together with statistical modeling, machine learning (ML) and deep learning.
- NLP also plays a growing role in enterprise solutions that help streamline and automate business operations, increase employee productivity and simplify mission-critical business processes

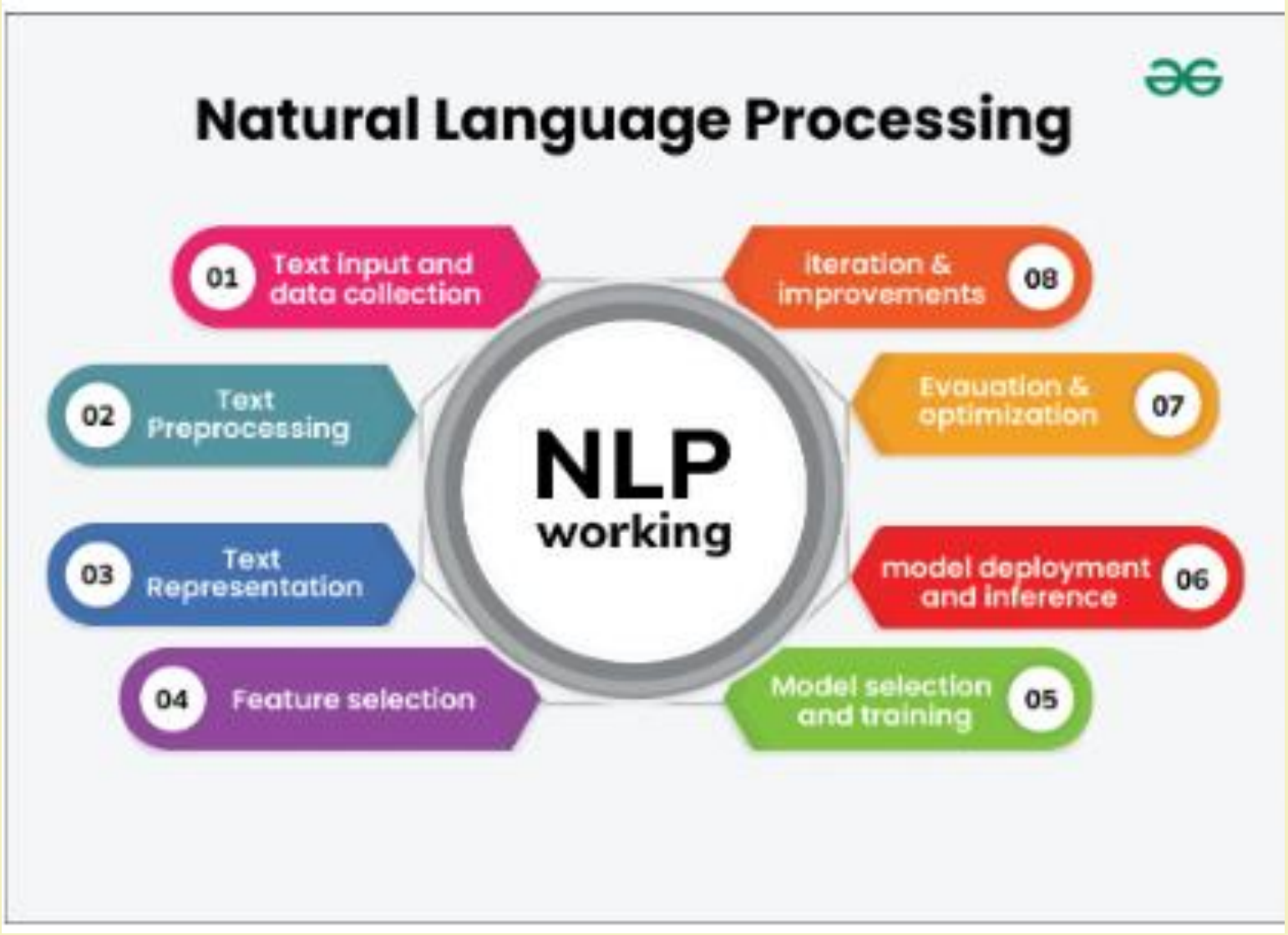
Benefits of NLP

- **Faster insight discovery**:
  - Organizations can find hidden patterns, trends and relationships between different pieces of content. Text data retrieval supports deeper insights and analysis, enabling better-informed decision-making and surfacing new business ideas.
- **Greater budget savings**:
  - With the massive volume of unstructured text data available, NLP can be used to automate the gathering, processing and organization of information with less manual effort.
- **Quick access to corporate data**:
  - An enterprise can build a knowledge base of organizational information to be efficiently accessed with AI search. For sales representatives, NLP can help quickly return relevant information, to improve customer service and help close sales.

Challenges of NLP

- **Biased training:** The more diverse the users of an NLP function, the more significant this risk becomes, such as in government services, healthcare and HR interactions. Training datasets scraped from the web, for example, are prone to bias.
- **Misinterpretation**: NLP solutions might become confused if spoken input is in an obscure dialect, mumbled, too full of slang, homonyms, incorrect grammar, idioms, fragments, mispronunciations, contractions or recorded with too much background noise.
- **New vocabulary:** New words are continually being invented or imported. The conventions of grammar can evolve or be intentionally broken. In these cases, NLP can either make a best guess or admit it's unsure—and either way, this creates a complication.
- **Tone of voice**: When people speak, their verbal delivery or even body language can give an entirely different meaning than the words alone. Exaggeration for effect, stressing words for importance or sarcasm can be confused by NLP, making the semantic analysis more difficult and less reliable.

How NLP Works



The Basics: How NLP is used in Robotics

- Traditionally, robots only understood coordinates (e.g., "Move arm to X:10, Y:5"). NLP changes this into a three-step process:

Step A: Understanding Intent: The robot uses an LLM (like GPT-4 or Llama) to turn "Make me a coffee" into a list of logical steps (find cup, move to machine, press button).

Step B: Grounding: The robot must connect words to the real world. If you say "Pick up the red mug," NLP helps the robot identify which object in its camera feed matches the label "red mug."

Step C: Feedback Loop: If the robot fails, you can give verbal corrections like "Higher" or "To the left," which the NLP system translates into adjustments.

Vision-Language-Action (VLA)

- Vision-Language-Action (VLA) models are AI systems that bridge the gap between perception and physical movement, allowing robots to understand natural language instructions and visual cues to perform tasks. OpenVLA and TinyVLA represent two different approaches to this technology, focusing on scale and efficiency, respectively.

- How a VLA Works: Imagine you tell a robot arm: "Put the banana in the green bowl."

1. Tokenization: The VLA turns your sentence into "tokens" (numbers the computer understands).
2. Vision Encoding: The robot's camera takes a picture. A "Vision Transformer" turns that image into a grid of visual features.
3. The "VLA" Fusion: The model overlays the text tokens on top of the image features. It "sees" the banana and the green bowl specifically because the text told it what to look for.
4. Direct Action Output: Instead of saying "I see a banana," the model's next "word" is actually a set of numbers representing Robot Actions (e.g.[0.1, -0.2, 0.5, 0.0])
5. Execution: These numbers go straight to the robot's motors, moving the hand toward the banana

Key Ethical Concerns

- **Accountability and Liability:** As robots become more autonomous, determining who is responsible for their actions—the manufacturer, the programmer, or the owner—becomes legally and ethically challenging.
- **Algorithmic Bias and Fairness:** AI-driven robots can inherit and amplify biases from their training data, leading to discriminatory outcomes in sensitive fields like healthcare, hiring, and law enforcement.
- **Transparency and Explainability:** Many advanced AI systems are "black boxes," making it difficult for humans to understand why a robot made a specific decision, which undermines public trust.
- **Safety and Reliability:** Ensuring that autonomous systems operate within strictly defined safety limits is critical to preventing physical harm to humans and property.

Major Societal Implications

- **Workforce Transformation:**

1. **Job Displacement:** Automation threatens millions of jobs, particularly in manufacturing, logistics, and customer service.

2. **New Opportunities:** Conversely, it creates demand for high-skilled roles in AI development, maintenance, and oversight.

- **Human-Robot Interaction:**

1. **Emotional Attachment:** Humans naturally anthropomorphize robots, leading to ethical concerns regarding deception and the replacement of genuine human companionship in caregiving roles.

2. **Psychological Impact:** Working closely with robots can sometimes lead to feelings of isolation, loss of autonomy, or decreased job satisfaction among human workers.

- **Privacy and Surveillance:** Robots equipped with advanced sensors and cameras can collect vast amounts of personal data, raising significant risks of unauthorized surveillance and data misuse.

- **Military and Security:** The development of Lethal Autonomous Weapon Systems (LAWS), or "killer robots," sparks intense international debate over delegating life-and-death decisions to machines.

Thank You